

EAST SALE, Australia

WMO: 949070

Lat: 38.1156S

Lon: 147.1322E

Elev: 27

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.3	34.4	30.6	24.8	42.0	32.2	26.7	42.2	30.7	53.8	27.5	53.8	4.4	310	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	91.6	67.0	86.1	66.5	81.2	65.4	70.5	83.0	68.9	79.8	67.4	77.0	11.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.1	73.1	65.6	94.5	71.9	64.1	89.6	70.9	34.4	83.3	33.1	80.5	31.9	77.0	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.3	23.9	21.1	DB	27.5	102.9	2.0	4.0	26.0	105.7	24.9	108.0	23.8	110.3	22.3	113.2
			WB	27.8	74.4	2.2	1.9	26.2	75.8	24.9	76.9	23.6	77.9	22.0	79.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	67.0	66.7	63.5	57.8	53.0	48.8	47.9	49.4	52.6	56.2	60.1	63.5	(6)
	DBStd	8.27	6.19	5.59	5.48	4.67	3.94	3.74	3.61	3.74	4.43	5.13	5.60	5.91	(7)
	HDD50	246	0	0	0	2	14	67	85	56	18	4	0	0	(8)
	HDD65	3202	45	40	97	222	372	487	530	485	373	280	169	102	(9)
	CDD50	2853	527	468	418	235	107	29	20	36	96	195	305	417	(10)
	CDD65	333	107	88	49	5	0	0	0	0	1	5	24	54	(11)
	CDH74	3295	986	695	444	87	5	0	0	0	26	153	330	569	(12)
	CDH80	1314	452	276	162	19	0	0	0	0	3	42	121	239	(13)
(14) Wind	WSAvg	8.2	9.1	8.6	8.1	7.0	7.6	7.3	7.8	8.3	8.6	8.5	9.2	8.8	(14)
(15) Precipitation	PrecAvg	22.9	1.7	1.5	1.8	1.9	2.0	1.8	1.7	1.7	2.0	2.0	2.4	2.2	(15)
	PrecMax	34.9	4.9	6.5	6.1	5.4	7.5	7.7	6.6	4.4	8.1	5.1	5.5	6.0	(16)
	PrecMin	11.9	0.0	0.0	0.2	0.3	0.4	0.2	0.3	0.2	0.5	0.2	0.6	0.1	(17)
	PrecStd	5.4	1.1	1.3	1.3	1.3	1.7	1.5	1.2	0.8	1.1	1.0	1.3	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.7	96.0	92.3	81.8	72.9	65.5	65.2	69.8	77.5	85.4	90.5	96.4	(19)
		MCWB	70.0	69.0	66.6	62.5	57.8	54.7	52.6	55.1	58.5	63.0	65.7	67.5	(20)
	2%	DB	91.2	88.2	84.1	75.6	68.1	61.2	61.0	64.4	71.6	78.0	82.8	86.8	(21)
		MCWB	68.1	67.4	65.9	61.3	57.1	52.5	50.9	52.9	56.6	60.4	64.2	66.4	(22)
	5%	DB	84.2	81.9	78.5	70.7	64.5	58.9	58.4	61.1	66.6	72.5	76.7	79.8	(23)
		MCWB	67.7	67.5	64.9	59.7	55.4	51.8	50.1	51.4	55.4	58.9	62.4	65.0	(24)
	10%	DB	78.2	77.0	73.7	67.3	61.8	56.9	56.2	58.4	62.7	67.1	71.2	74.6	(25)
		MCWB	65.7	65.9	63.3	58.4	54.1	50.9	49.4	50.0	53.6	57.2	60.8	63.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.3	72.6	70.4	65.2	61.0	57.1	54.8	57.0	61.5	65.3	69.0	71.0	(27)
		MCDB	88.8	85.7	81.5	76.4	68.2	61.0	59.7	66.3	72.6	75.9	81.8	85.4	(28)
	2%	WB	70.8	70.2	68.0	63.0	58.4	54.7	52.8	54.5	58.6	62.2	65.8	68.3	(29)
		MCDB	84.2	81.1	77.2	70.6	64.9	59.1	57.8	61.4	66.9	73.6	77.7	80.1	(30)
	5%	WB	68.8	68.5	66.3	61.4	56.9	53.1	51.3	52.5	56.3	60.1	63.8	66.3	(31)
		MCDB	80.0	77.9	75.4	68.1	62.5	57.4	56.4	58.7	64.4	69.7	73.5	76.6	(32)
	10%	WB	67.0	67.0	64.6	59.7	55.2	51.6	50.1	51.0	54.4	58.0	61.9	64.4	(33)
		MCDB	76.7	75.0	72.5	65.7	60.4	55.7	55.3	56.9	61.4	66.1	69.7	73.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	20.1	20.3	19.9	18.3	17.4	17.3	18.6	20.2	20.9	20.3	20.8	(35)
		MCDBR	32.5	29.5	29.3	26.2	22.6	20.2	20.9	24.0	27.8	31.8	31.1	31.7	(36)
		MCWBR	12.1	11.8	12.4	13.2	13.3	12.9	12.8	14.1	15.3	15.6	13.7	13.1	(37)
	5% WB	MCDBR	26.7	23.7	23.8	21.9	21.0	18.5	18.1	21.0	24.4	27.2	26.4	26.3	(38)
		MCWBR	11.3	10.7	10.9	11.6	13.1	12.8	12.0	13.3	14.4	14.4	12.6	12.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.384	0.369	0.341	0.333	0.312	0.301	0.303	0.310	0.334	0.344	0.352	0.380	(40)
	taud		2.415	2.482	2.551	2.549	2.583	2.605	2.586	2.563	2.493	2.490	2.496	2.413	(41)
	Ebn at Noon		301	298	294	276	264	258	264	280	291	303	309	304	(42)
	Edh at Noon		39	35	30	26	22	20	22	25	31	34	36	39	(43)
(44) All-Sky Solar Radiation	RadAvg		2167	1863	1523	1083	747	594	677	940	1312	1707	1973	2194	(44)
(45)	RadStd		137	117	99	68	38	41	35	41	94	115	126	129	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.59	N/A	N/A	N/A	+0.90	+1.24	N/A	-187	+247	+76	(46)
(47)	Regional (28 neighbors)	+0.86	N/A	N/A	+1.35	N/A	N/A	-25	-230	+304	+106	(47)

Nomenclature: See separate page